# A Transformer-Based Network for Hyperspectral Object Tracking

Long Gao, *Member, IEEE*, Langkun Chen, Pan Liu, Yan Jiang, Weiying Xie, *Senior Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*— With the abundant spectral information, the hyperspectral images could be benefit for tracking the target in various application scenarios. Most of the predominant hyperspectral object tracking methods were based on transferred red–green–blue (RGB) object tracking networks, since the lack of training data. Different strategies of processing the hyperspectral images to adapt the transferred RGB object tracking networks have been exploited. However, the existing strategies led to lose the spectral information or the interaction information between bands and have shown limited performances. In this article, a novel transformer-based hyperspectral object tracking algorithm (Trans-HST) is proposed to make advantages of the spectral information with transformer modules. In Trans-HST, the cross-band groups of feature enhancement (CBFE) is introduced to reduce the negative effects of the interaction information loss. To address the problem of spectral information loss, the transformer-based deep features' fusion (TDFF) fuses the deep features corresponding to different groups of bands in the hyperspectral images and integrates the deep feature corresponding to the original hyperspectral images into the fused features. Experiments on the commonly used hyperspectral object tracking dataset have been applied to verify the effectiveness of the two proposed modules, and they indicate the superior performance of the Trans-HST comparing with other RGB and hyperspectral trackers.**

***Index Terms*— Feature enhancement, feature fusion, hyperspectral object tracking, transformer.**

## I. Introduction

VISUAL object tracking has been a fundamental task in computer vision and has been applied in unmanned aerial vehicle (UAV)-borne surveillance, robot vision, unmanned driving, and other fields [1]. However, designing a robust tracking system is still challenging. With limited information of the target in the red–green–blue (RGB) images, an outstanding tracking model must be flexible to counter rapid target appearance changes and resilient to occlusion and background clutter [2], [3], [4], [5], [6]. Recently, with the advancement of the spectral imaging technology, hyperspectral videos (HSVs) can be obtained through HS snapshot sensors [7], [8]. It makes exploring the algorithms of tracking objects with HSV possible. Different from RGB images, the underlying material information in HSI can be applied for the tracker to distinguish the target from complex background [8]. As shown in Fig. 1, the tracker based on HSV [transformer-based hyperspectral object tracking algorithm (Trans-HST)] outperforms the other trackers by applying the spectral information against the background interference. Consequently, tracking object with HSV has attracted increasing attention, and some impressive explorations have been made in [9], [10], [11], [12], [13], [14], [15], [16], and [17].

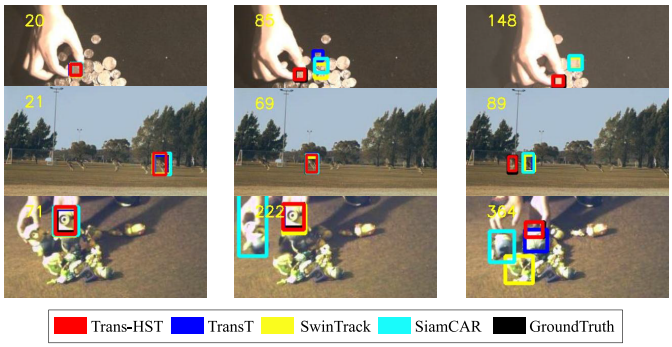


Fig. 1. Visualized comparisons of the Trans-HST and RGB trackers.

The prevailing HS tracking methods were investigated based on deep learning. Deep learning models needed large number of annotated training data to generate worthy discriminative ability. Acquiring HSV with annotations was expensive, and the HSV datasets were smaller than the commonly used large-scale datasets for RGB object tracking, i.e., GOT-10K [18], TrackingNet [19], and LaSOT [20]. Accordingly, HS trackers based on deep learning applied transferred RGB tracking model to design the tracking networks. Different from the three-channel RGB data, HSI available for object tracking contained more than 15 channels which was unapplicable for traditional 2-D tracking networks. There were two schemes for addressing the problem. First, researches converted HSI into a three-channel image and tracked the target with the tracking network pretrained with RGB data [8], [13], [14], [15], [16], [17]. Second, some works divided the HSI into multiple three-channel false-color images and fused the tracking results which were predicted with the three-channel false-color images [21], [22], [23]. For each

Manuscript received 10 April 2023; revised 10 September 2023; accepted 11 October 2023. Date of publication 17 October 2023; date of current version 25 October 2023. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant JB210103 and Grant CEIEC-2022-ZM02-0247 and in part by the Youth Innovation Team of Shaanxi Universities. *(Corresponding author: Weiying Xie.)*

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Digital Object Identifier 10.1109/TGRS.2023.3325049

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false-color image, the tracking results were obtained with transferred RGB tracking networks. However, the dimension reduction in the converting procedures led to the loss of spectral information and limits the performance improvement. Meanwhile, the dividing processes lead to the loss of the interaction information between bands. Furthermore, processing each false-color images with the whole tracking network, respectively, is inefficient, and the speed of the trackers is lower than 10 frames/s [21], [22], [23]. In general, how to achieve the benefit of the rich spectral information in HSV is still challenging for the HS object tracking algorithms based on deep learning.

In this article, the HS object tracking algorithm based on transformer, Trans-HST, is proposed. Trans-HST divides the HSI into multiple false-color images to suite the transferred RGB tracking networks and fuses the features corresponding to the false-color images for efficiency. First, the cross-band groups of feature enhancement (CBFE) is introduced to reduce the adverse effects generated by dividing the HSI into multiple false-color images. The CBFE reconstructs the features corresponding to the original HSI with deep features of the false-color images and learns the self-attention of the reconstructed features with a multihead self-attention (MSA) module. By learning the dependency information between channels and integrating it into the reconstructed feature map, each feature of the false-color image contained in the reconstructed feature map can be enhanced with spectral information of the other bands.

Second, the transformer-based deep features' fusion (TDFF) is proposed to fuse deep features extracted from false-color images, which reduce the spectral information loss. Different from the exiting methods which fuse the tracking results obtained with the transferred RGB tracking network, the Trans-HST fuses the deep features of the false-color images with the transformer-based network and tracks the target with the fused features. Since the fused features share the same size with one deep feature of the false-color image, the computational cost can be reduced. The skilled ways to fuse the deep features of the false-color images are summation or concatenation in the channel domain. As shown in the ablation study, the performance of the tracker with the two fusion methods is limited as the interaction information between channels in different false-color images is neglect in fusion procedure. Therefore, TDFF fuses the features with weighted summation and implements a multihead cross-attention module to augment interaction information across tokens into the fused features.

The main contributions of this article are summarized as follows.

1) The CBFE is introduced to magnify the discrimination ability of the features of the false-color images. The input of the module is a deep feature corresponding to the original HS image. The CBFE is constructed with an MSA module to learn the interaction information across the features extracted with different groups of bands and integrate the information into the outputs.
2) The TDFF is proposed to fuse the features of the false-color images. A multihead cross self-attention (MCSA) module is implemented on the TDFF to augment the concatenated features into the summation feature. The concatenated feature is obtained by concatenating the features of the false-color images in the channel domain. The summation feature is generated by fusing the five deep features corresponding to the false-color images with learnable weights.
3) Based on the CBFE and TDFF, a novel HS object tracking algorithm, Trans-HST, is proposed. The feature extraction network and head network are initialized with the parameters of the tracking network trained with RGB data, and further trained along with the CBFE and TDFF using HS data. The validities of the two proposed modules are verified in validation study, and experiments on the commonly used HS object tracking benchmark indicate the superiority in performance compared with other state-of-the-art methods.

The rest of this article is organized as follows. Section II briefly reviews the basic theories related to trackers based on RGB data and HS data, respectively. Furthermore, the transformer mechanism is reviewed because the main contributions of this article are based on the transformer. In Section III, the proposed Trans-HST is presented in detail. In Section IV, the experimental results and comprehensive analysis are illustrated before concluding in Section V.

## II. Related Work

Hyperspectral object tracking has thrived in recent years due to the ability to differentiate materials and can be applied in remote sensing, i.e., traffic management [24], route determination [25], and sea ice detection [26]. Most of the recent HS object tracking algorithms were based on RGB object tracking algorithms, and the transformer method is applied in the proposed algorithm. Therefore, brief reviews on visual object tracking in RGB data and HS data are presented, respectively. Then, the review of the transformer is provided.

### A. RGB Object Tracking

Recent prevailing tracking methods, Siamese-based algorithms, formulated the tracking task as a matching problem, which computed the similarity matrix between a template and search regions [2], [4], [27], [28], [29], [30]. SiamFC learned the similarity matrix of the features of the target patch and the search patch with a fully convolution Siamese network [27]. Based on SiamFC, a region proposal network (RPN) [31] has been introduced in the Siamese network to perform one-shot detection on the similarity matrix, and it significantly improved the performance [2], [29]. The RPN was sensitive to hyperparameters, i.e., numbers, sizes, and aspect ratios of the anchor boxes. To dispel the limitation of the RPN, anchor-free methods have been implemented in the Siamese networks, which estimated the state of the target directly with the classification and regression score maps [4], [30]. Moreover, the attention modules have been implemented in the Siamese networks to enhance the discriminative ability of the features [32], [33], [34], [35]. The Siamese-based algorithms have achieved considerable performance improvement for the

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RGB object tracking task. The discrimination ability learned from large numbers of RGB training data can be beneficial for the trackers based on HSV.

### B. Hyperspectral Object Tracking

Different from the RGB videos, HSV contains both spectral and spatial information, and the enrichment spectral information is beneficial for the discrimination ability of the tracker due to the underlying material information. Recently, increasing attention has been drawn to explore robust trackers based on HSV.

The pioneer works in HS object tracking were based on handcrafted features [7], [9]. Xiong et al. [7] proposed two handcrafted feature extractors to obtain the material information in an HSI and fed the features into the background-aware correlation filter to track the targets in HSV. More research studies were based on deep learning methods, which could be categorized into three mainstreams. First, researchers explored different methods to reduce the dimension of HSI and tracked the target with transferred tracking network which was pretrained with RGB data [11], [13], [14], [15], [36], [37]. Some algorithms applied the bands' selection modules based on the traditional methods to convert the HSI into three-channel images, which adapted the trackers based on RGB data [11], [37]. Meanwhile, specially designed networks were applied in the front of tracking networks to diminish the channels of the HSI and tracked the target with the RGB tracking network [14], [15], [36]. Second, the trackers regrouped the bands according to different criteria to form the several false-color images and computed the score maps with the transferred tracking network on the false-color images, respectively [16], [21], [22], [23]. The states of the targets were estimated by fusing the score maps. Third, instead of transferring the network pretrained with RGB data, some works proposed newly designed networks to extract the features of the HSI images [38], [39], [40]. 3-D convolution layers were introduced to directly extract deep features of the HSI [38], [39].

### C. Transformer

Transformer was first proposed in the field of natural language processing, and then introduced into the field of computer vision [5], [41]. The preliminary work [41] proposed self-attention blocks to capture the long-range interaction information. The encoder–decoder based on self-attention was introduced to the detection task and achieved obvious performance improvement [42], [43]. Instead of integrating the transformer blocks in convolution neural networks, vision transformer (ViT) first built the network with transformer blocks [44]. To improve the efficiency, works in [45] and [46] reduced the number of the tokens by inserting spatial reduction or convolution layers before the self-attention module to merge tokens. Moreover, a window-based transformer has been introduced to improve the efficiency with calculating the self-attention within small regions [47].

Transformer mechanism was applied in the object tracking task [5], [6], [35], [48], [49], [50]. For the RGB images, TransT fused the deep features of the template and search area with modules based on self-attention [5]. Moreover, researchers exploited the methods which implemented the encoder and the decoder from the transformer in the tracking networks and gained considerable improvement [35], [48], [49]. Furthermore, [6] and [50] introduced the transformer-based feature extraction network and built the tracking network in all attention manner. For the HSI, the transformer mechanism has also been applied. The transformer was implemented to aggregate the information extracted from the HSI and that from RGB images to gain robust performance [38].

## III. Proposal Algorithm

In this section, the pipeline of Trans-HST is introduced in detail. Then, the CBFE module which enhances the deep features corresponding to the false-color images is presented. In addition, the TDFF module which fuses the features corresponding to the false-color images is exhibited. Finally, the training and inference procedures of the proposed method are described.

### A. Architecture of the Transformer-Based Network

*1) Feature Extraction:* As shown in Fig. 2, the network takes two images as inputs, the template image and the search image which shares location and is twice the size of the template. The Trans-HST network divides the high-dimensional template image and search image into groups of three-channel false-color images for feature extraction. Specifically, the search image, $\mathbf{X} \in \mathbb{R}^{H\times W\times 16}$, is separated into five false-color images, $\mathbf{x}_i \in \mathbb{R}^{H\times W\times 3}$, and the five false-color images form the search group $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}$. The 16th bands in the HSI are removed, and the remaining channels are separated sequentially. Thus, they can be divided into five groups, and each of them has three channels, i.e., ($\{1, 2, 3\}$, $\{4, 5,6\}$, $\{7, 8, 9\}$, $\{10, 11, 12\}$, $\{13, 14, 15\}$). The heights of the search images and template images are defined as $H$ and $h$, and the widths of the patches are defined as $W$ and $w$, respectively. The false-color images are fed into convolution neural networks, which take a three-channel image as input to extract deep features. The same operations are performed on the template image $\mathbf{Z} \in \mathbb{R}^{h\times w\times 16}$ to obtain the five false-color images $\mathbf{z}_i \in \mathbb{R}^{h\times w\times 3}$ and form the template group $\{\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4, \mathbf{z}_5\}$. The feature extraction networks are modified ResNet-50 [51] and share the weights. The outputs of the feature extraction networks are $\mathbf{F}_{x_i} \in \mathbb{R}^{d\times(H/8)\times(W/8)}$ corresponding to search group and $\mathbf{F}_{z_i} \in \mathbb{R}^{d\times(h/8)\times(w/8)}$ corresponding to the template group, where $d$ is the number of channels.

*2) Multiple Deep Features' Fusion:* After feature extraction, two transformer-based fusion branches are implemented for strengthening and fusing the two groups of the deep features, i.e., $\mathbf{F}_{x_i} \in \mathbb{R}^{d\times(H/8)\times(W/8)}$ and $\mathbf{F}_{z_i} \in \mathbb{R}^{d\times(h/8)\times(w/8)}$, respectively. Each transformer-based fusion branch consists of two sequentially implemented modules, CBFE and TDFF. The division strategy in feature extraction leads to loss of interaction information between bands. Hence, the CBFE reforms the divided features into one united feature which is corresponding to the original HS image and learns the self-attention of

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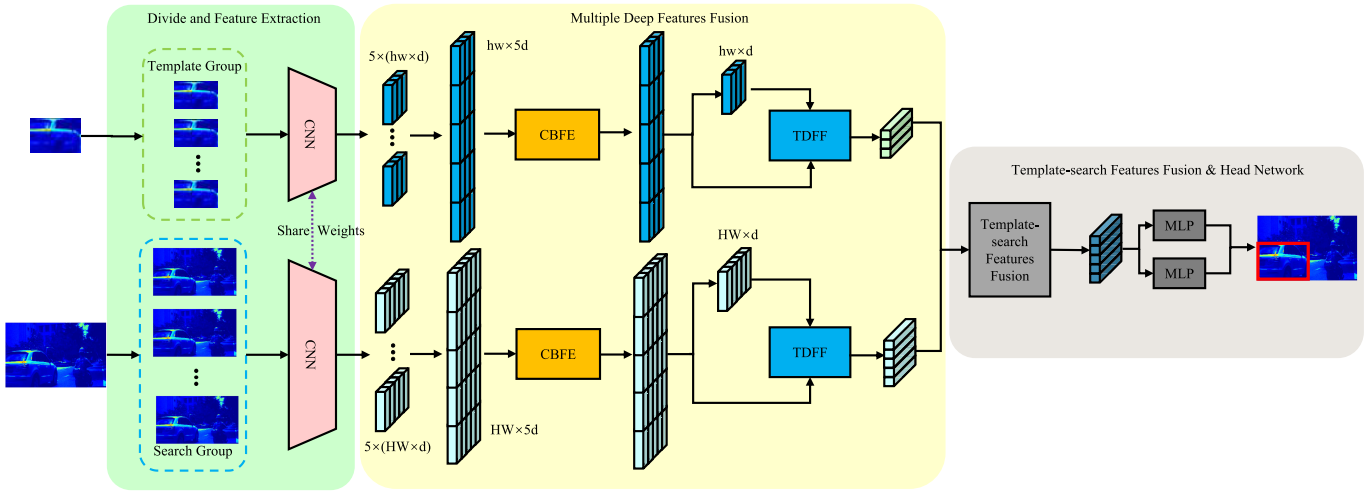


Fig. 2. Pipeline of Trans-HST. The feature extraction, template-search feature fusion, and head network are constructed and initialized with the transferred RGB tracking network.

the united feature for reducing the influence of the loss of interaction information. The inputs of the CBFE are a set of vectors obtained by concatenating the flattened features, i.e., $\mathbf{F}'_{x_i} \in \mathbb{R}^{d\times(HW/64)}$ or $\mathbf{F}'_{z_i} \in \mathbb{R}^{d\times(hw/64)}$, from the same group. An MSA module is implemented in the CBFE to enhance the fused features with long-range interaction information among tokens from the five deep features. The outputs of the CBFE are $\tilde{\mathbf{F}}'_X \in \mathbb{R}^{5d\times(HW/64)}$ and $\tilde{\mathbf{F}}'_Z \in \mathbb{R}^{5d\times(hw/64)}$.

The TDFF is designed for reducing the size of the deep features of the template and search patch. Directly sending the five deep features into the template-search features' fusion network and the head network costs high computational expense. The TDFF fuses the five vectors, i.e., $\mathbf{F}_{x_i} \in \mathbb{R}^{d\times(HW/64)}$ or $\mathbf{F}_{z_i} \in \mathbb{R}^{d\times(hw/64)}$, into one feature map, $\tilde{\mathbf{F}}''_X \in \mathbb{R}^{d\times(HW/64)}$ or $\tilde{\mathbf{F}}''_Z \in \mathbb{R}^{d\times(hw/64)}$. The TDFF takes two feature maps as input. One is the output of the CBFE, i.e., $\tilde{\mathbf{F}}'_X \in \mathbb{R}^{5d\times(HW/64)}$ or $\tilde{\mathbf{F}}'_Z \in \mathbb{R}^{5d\times(hw/64)}$. The other one is a feature map which is acquired by adding the five features with learnable weights. The five features are obtained by separating $\tilde{\mathbf{F}}'_X \in \mathbb{R}^{5d\times(HW/64)}$ and $\tilde{\mathbf{F}}'_Z \in \mathbb{R}^{5d\times(hw/64)}$.

*3) Template-Search Features' Fusion and Head Network:* The template-search features' fusion network for integrating the features of the template and search image is similar to TransT [5]. Two modules, ego-context augment (ECA) modules and cross-feature augment (CFA) modules, are implemented in the network. ECA enhances the representation of the input with MSA, and CFA fuses the features corresponding to the template and search image. The head network has two branches, classification branch and regression branch, and each branch includes three convolution layers and a ReLU active function. The classification branch outputs a score map, which indicates the possibility of each sample is the target. The regression branch outputs a coordinate map to estimate the bounding box of the target.

### B. Cross-Band Groups of Feature Enhancement

For extracting the features adapting to the HS tracking task while maintaining the discrimination ability learned from the RGB data, the HSI are divided into five false-color images.

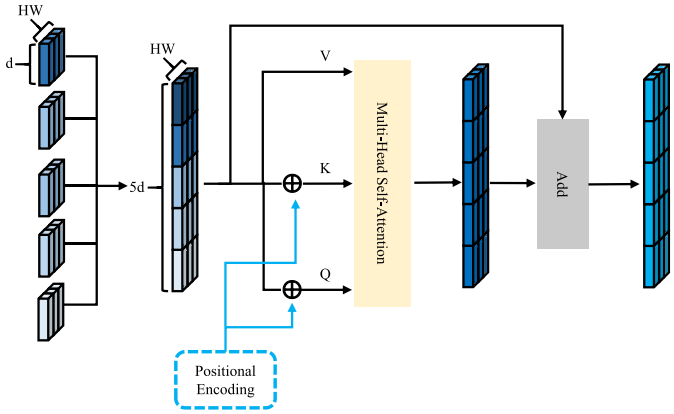


Fig. 3. Structure of the cross-band groups of feature enhancement module.

The strategy loses some interaction information between bands of the HSI. Therefore, the CBFE learns the long-range interaction information between channels in the five features and enhances the representation abilities of the five features with the learned interaction information. As shown in Fig. 3, the CBFE rebuilds the deep features corresponding to the original HSI with the features extracted from the false-color images and applies the MSA learning module to learn the long-range interaction between channels from all the five deep features.

Taking the features, $\mathbf{F}'_{x_i} \in \mathbb{R}^{d\times(HW/64)}$, which are corresponding to the false-color image in the search group as an example, CBFE rebuilds the deep feature corresponding to the original HS image by concatenating the features in the channel domain. The concatenated feature is $\mathbf{F}'_X \in \mathbb{R}^{5d\times(HW/64)}$. In CBFE, the MSA computes the self-attention over the united feature in the channel domain. The input of MSA is projected into queries ($\mathbf{Q}_i$), keys ($\mathbf{K}_i$), and values ($\mathbf{V}_i$) as follows:

$$\begin{cases} \mathbf{Q}_i = \mathbf{F}'_X \mathbf{W}_i^Q \\ \mathbf{K}_i = \mathbf{F}'_X \mathbf{W}_i^K \\ \mathbf{V}_i = \mathbf{F}'_X \mathbf{W}_i^V \end{cases} \tag{1}$$

where $\mathbf{W}_i^Q \in \mathbb{R}^{(HW/64)\times d_k}$, $\mathbf{W}_i^K \in \mathbb{R}^{(HW/64)\times d_k}$, and $\mathbf{W}_i^V \in \mathbb{R}^{(HW/64)\times d_k}$ are the parameters of the linear projections in the

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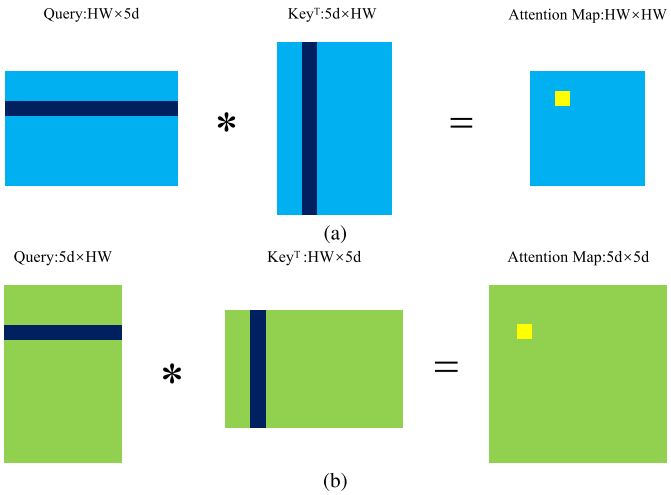


Fig. 4. Illustration of computing the self-attention map in different domains. (a) Computing the self-attention in the space domain. (b) Computing the self-attention in the channel domain.

$i$th matrices, and $d_k$ represents the size of $\mathbf{K}_i$ respectively. Different from the original MSA module which computes the self-attention of the input in the space domain, the MSA in the CBFE computes that in the channel domain. As Fig. 4(a) shows, the original MSA module captures the long-range dependency information between vectors corresponding to the pixels of the deep feature. The MSA module in the CBFE captures the long-range dependency information between channels corresponding to the bands of the HSI as shown in Fig. 4(b). The mechanism of MSA can be summarized as follows:

$$\begin{cases} \text{head}_i = \text{softmax}\left(\dfrac{\mathbf{Q}_i\mathbf{K}_i^T}{\sqrt{d_k}}\right)\mathbf{V}_i \\ \tilde{\mathbf{F}}'_X = \mathbf{F}'_X + \text{concat}(\text{head}_1, \text{head}_2, \ldots, \text{head}_n)W^O \end{cases} \tag{2}$$

where $W^O \in \mathbb{R}^{nd_k\times(HW/64)}$ aret= the parameters of the linear projection, and $n$ is the number of heads. softmax$(\cdot)$ is the active function. concat$(\cdot)$ is the concatenation operation. In this work, $n = 8$, $d_k = 32$, and $(HW/64) = 256$ are the default values.

The self-attention learning has no ability to distinguish the position information of the input features. Therefore, a spatial positional encoding is generated with a sine function [5] and is augmented into the query and the key. In our case, the positional encoding is corresponding to the channels. The attention learning is reformed as follows:

$$\tilde{\mathbf{F}}'_X = \mathbf{F}'_X + \text{MSA}\left(\mathbf{F}'_X + \mathbf{P}, \mathbf{F}'_X + \mathbf{P}, \mathbf{F}'_X\right) \tag{3}$$

where $\mathbf{P}$ represents the positional encoding corresponding to the channels of $\mathbf{F}'_X$, and MSA$(\cdot)$ is the mutihead self-attention function.

### C. Transformer-Based Deep Features' Fusion

The TDFF is designed for fusing the deep features extracted from the three-channel false-color images. Fusing the tracking results of the weak trackers which track the target with deep features corresponding to the false-color images, respectively, is inefficient. Specifically, the five false-color images are processed with the whole tracking networks, which leads to the calculate speed being one-fifth of that of tracking the target with one RGB tracking network. Furthermore, each tracker which tracks the target with three-channel false-color images does not take advantage of the rich spectral information of HSI. On the contrary, Trans-HST fuses the deep features corresponding to false-color images and tracking the target with the fused features. Only one fused feature is processed by the fusion network and head network in the tracking network, and the fused feature is corresponding to the original HSI. As a result, the proposed TDFF improves the performance of the tracker considerably and runs at faster speed than that of the algorithms which fuse the tracking results of weak trackers. Fig. 5 illustrates the mechanism of the TDFF module.

As shown in Fig. 2, the inputs of the TDFF module are the outputs of the CBFE, $\tilde{\mathbf{F}}'_X \in \mathbb{R}^{5d\times(HW/64)}$, and $\mathbf{F}''_X \in \mathbb{R}^{d\times(HW/64)}$. $\mathbf{F}''_X$ is the acquired by calculating the weighted sum of five features as follows:

$$\mathbf{F}''_X = \sum_{i=1}^{n} w_i\mathbf{D}_{x_i} \tag{4}$$

where $w_i$ is a learnable weight, $n$ is the number of the features which is set to 5 in our case, and $\mathbf{D}_{x_i} \in \mathbb{R}^{d\times(HW/64)}$ is the feature which is obtained by dividing $\tilde{\mathbf{F}}'_X$ as follows:

$$\mathbf{D}_{x_i} = \tilde{\mathbf{F}}'_X[(i-1)d : (i\times d - 1), :]. \tag{5}$$

The TDFF consists of an MCSA. The inputs of the MCSA are queries $(\mathbf{q}_i)$, keys $(\mathbf{k}_i)$, and values $(\mathbf{v}_i)$ and can be obtained with the following equation:

$$\begin{cases} \mathbf{q}_i = \mathbf{F}''_X\mathbf{W}_i^q \\ \mathbf{k}_i = \tilde{\mathbf{F}}'_X\mathbf{W}_i^k \\ \mathbf{v}_i = \tilde{\mathbf{F}}'_X\mathbf{W}_i^v \end{cases} \tag{6}$$

where $\mathbf{W}_i^q \in \mathbb{R}^{(HW/64)\times d_k}$, $\mathbf{W}_i^k \in \mathbb{R}^{(HW/64)\times d_k}$, and $\mathbf{W}_i^v \in \mathbb{R}^{(HW/64)\times d_k}$ are the parameter matrices of the linear projection in the $i$th matrices. The MSA is calculated with (2) in the channel domain. Because the numbers of channel in the queries are different to that of the keys, two positional encodings which are corresponding to $\tilde{\mathbf{F}}'_X \in \mathbb{R}^{5d\times(HW/64)}$ and $\mathbf{F}''_X \in \mathbb{R}^{d\times(HW/64)}$ are applied in the self-attention learning procedure. The learning procedure can be formulated as follows:

$$\tilde{\mathbf{F}}''_X = \mathbf{F}''_X + \text{MCSA}\left(\mathbf{F}''_X + \mathbf{P}_1, \tilde{\mathbf{F}}'_X + \mathbf{P}_2, \tilde{\mathbf{F}}'_X\right) \tag{7}$$

where $\tilde{\mathbf{F}}''_X \in \mathbb{R}^{d\times(HW/64)}$ represents the output of the MCSA, and $\mathbf{P}_1 \in \mathbb{R}^{d\times(HW/64)}$ and $\mathbf{P}_2 \in \mathbb{R}^{5d\times(HW/64)}$ are the two positional encodings with different sizes because the input sizes of MCSA modules are different. MCSA$(\cdot)$ refers to the MCSA module.

### D. Tracking With Trans-HST

*1) Training:* The HS data for training HS tracking networks are insufficient, which may lead to overfitting. On the contrary, the amount of annotated RGB data for training the RGB trackers, i.e., the amount of training data for TransT, is more than 1.4 TB. Building and initializing the HS tracking network with transferred RGB network can retain the discriminative ability learned from the RGB training data. Therefore, the structures of the backbone, the template-search features' fusion

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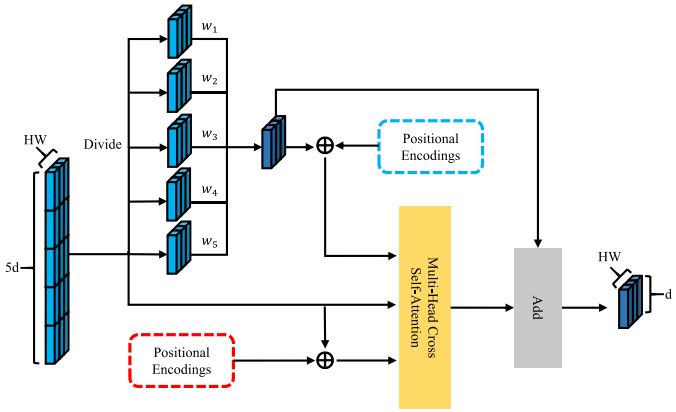


Fig. 5. Structure of the TDFF module.

network, and the head network of the Trans-HST are the same as TransT. The parameters of the three parts are initialized with the parameters in the TransT network which is pretrained with RGB data. The binary cross-entropy is implemented as the loss function for classification. For bounding box regression branch, the generalized IoU loss [52] is used and formulated as follows:

$$L_{\text{reg}} = \sum_i 1_{\{\text{IoU}(\mathbf{b}^i, \hat{\mathbf{b}}^i)\}} \left[ p L_{\text{GIoU}}\left(\mathbf{b}^i, \hat{\mathbf{b}}^i\right)\right] \tag{8}$$

where $L_{\text{reg}}$ is the generalized IoU loss function. $p$ is the weight parameter which highlights the samples with high classification score.

*2) Tracking:* The template is obtained in the first images of the sequence, and the search images are taken from the follow-up images. The template and search images are divided into five false-color images by grouping adjacent bands sequentially based on their indices. The location of the targets is estimated with the classification score maps outputted by the classification branch. The classification score maps include two channels, which represent the possibilities of the samples being the target or the background. The bounding box is calculated with the regression maps outputted from the regression branch. The regression map includes four channels, which represents the normalized coordinates. A penalty matrix based on the Hamming window is implemented on the classification score map as the other RGB trackers, and the weight value of the penalty matrix is set to 0.49 [5], [29].

## IV. Experiments

In this section, the experimental setups are first introduced. Second, an ablation study is conducted for verifying the effectiveness of the proposed modules. Third, Trans-HST is evaluated on popular HS object tracking benchmark with other HS object tracking algorithms.

### A. Experimental Setup

*1) Dataset:* The training and testing HS data are provided in [8], which are the first public and the most used hyperspectral object tracking dataset. The dataset includes HSV, RGB videos, and false-color videos which describe the same scenes for evaluation of trackers based on HSV and RGB videos. The HSV in the dataset consists of 16 bands which cover the wavelength from 470 to 620 nm. The scenes are categorized according to the 11 challenging facts [53]. There are 40 sequences and 35 sequences in the dataset prepared for training and testing, respectively. The area under the curve (AUC) of the success plot and the precision score at the threshold of 20 pixels (DP@20P) are applied as evaluation metrics in the experiments, and the two metrics are tested in one-pass evaluation (OPE) manner [53].

*2) Implementation Details:* The Trans-HST is instantiated in Python with PyTorch and trained on two RTX 3090 cards. The network for initializing is pretrained on the RGB datasets, and the details of the pretraining procedure can be found in [5]. In the training procedure on the HSV, the parameters initialized with the pretrained network are frozen according to [54]. The batch size is set to 8, and a total of 100 epochs are performed using AdamW [55]. The parameters' learning rate is set to 1e−6, with a weight decay of 1e−5. The learning rate decreases by factor 10 after 60 epochs. The training set is enlarged with common data augmentation, i.e., horizontal flip and brightness jittering. The sizes of the inputs, i.e., template images and search images, are resized to 128 × 128 and 256 × 256, respectively. The backbone is the modified ResNet-50 which removes the fifth block and modifies the convolution stride of the downsampling unit from 2 to 1 for resolution increasing. To divide the HSI into five false-color images, band 16 of the HSI is removed. The spatial sizes of the deep features of the false-color images corresponding to the template images and search images are 16 × 16 and 32 × 32, respectively, and the channel of each deep features of the false-color images is 1024. The numbers of heads in the MSA and MCSA are set to 8.

### B. Ablation Study

To validate effectiveness of the two proposed modules, an experiment on the HSV dataset is performed, and the results are shown in Table I. In the experiment, the baseline and weighted feature fusion represent the trackers, which fuse the features with additive fusion and learnable weights $w_i$, respectively. The concatenated represents the tracker, which fuses the features by concatenating the features in the channel domain and reducing the dimension with a convolution layer. The CBFE-Space and CBFE-Channel represent the trackers compute the MSA with the methods in Fig. 4(a) and (b), respectively, and fuse the features as the method described in (4). TDFF represents the tracker fuses the features without CBFE.

Compared with the baseline, fusing the features with learnable weights shows better performance because the features corresponding to false-color images contribute to tracking the target differently. The tracker, concatenated, performs worse than the baseline since the dimension reduction causes information loss. Based on the weighted feature fusion, CBFE-Space and CBFE-Channel improve the performances of the trackers by 6.7% and 7.6% in AUC scores, respectively. CBFE-Channel achieves better performance than the CBFE-Space. Because the interaction information loss is

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TABLE I
ABLATION STUDY OF THE PROPOSED TRACKER ON THE HS OBJECT TRACKING DATASET

| | AUC | ΔAUC% | DP@20P | Δ DP@20P% |
|---|---|---|---|---|
| Baseline | 0.586 | - | 0.820 | - |
| Weighted Feature Fusion | 0.633 | 4.7% | 0.862 | 4.2% |
| Concatenated | 0.548 | -3.8% | 0.801 | -1.9% |
| CBFE-Space | 0.653 | 6.7% | 0.874 | 5.4% |
| CBFE-Channel | 0.662 | 7.6% | 0.885 | 6.5% |
| TDFF | 0.666 | 8.0% | 0.897 | 7.7% |
| CBFE+TDFF | 0.682 | 9.5% | 0.921 | 10.1% |

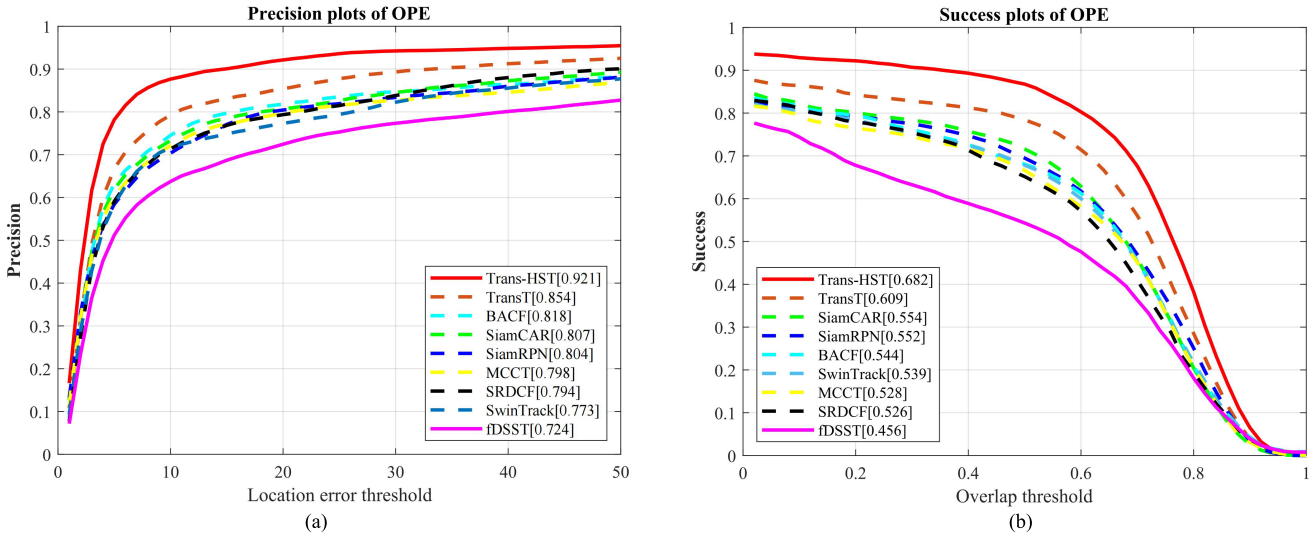


Fig. 6. Comparisons of Trans-HST and trackers on the corresponding false-color videos. (a) Precision plots. (b) Success plots.

TABLE II
PERFORMANCES OF TRACKERS WITH DIFFERENT BANDS OF REGROUPING STRATEGIES

| Bands Regrouping Strategies | AUC | DP@20P |
|---|---|---|
| {0,3,6},{1,4,7},{2,5,8},{9,11,13},{10,12,14} | 0.671 | 0.907 |
| {0,5,10},{1,6,11},{2,7,12},{3,8,13},{4,9,14} | 0.666 | 0.906 |
| {0,3,4},{1,12,13},{2,8,11},{5,7,10},{6,9,14} | 0.675 | 0.91 |
| {0,1,2},{3,4,5},{6,7,8},{9,10,11},{12,13,14} | 0.682 | 0.921 |

caused by dividing the HSI in the channel domain, learning the long-range dependency information across the channels (CBFE-Channel) reduces the impact of the information loss more directly than learning long-range dependency information across the vectors (CBFE-Space). The TDFF outperforms the weighted feature fusion because it augmented the feature maps corresponding to the original HSI into the fused features. The tracker equipped with both CBFE and TDFF shows the best performance in this experiment. Compared with the baseline, it gains performance improvement by 9.5% in AUC score and 10.1% in DP@20P score.

The performances of trackers with different band regroup strategies are evaluated on the HS dataset and reported in Table II. The first two lines are regrouping the bands with gaps of 3 and 5, respectively. The third line is regrouping the bands randomly, and the fourth line is regrouping the bands sequentially. It should be noted that regrouping the bands sequentially achieves the best performance. Therefore, the proposed tracker divided the HS images by sequentially regrouping the bands.

### *C. Quantitative Comparison With the State-of-the-Art RGB Trackers*

In this experiment, the Trans-HST is evaluated with eight RGB trackers, i.e., TransT [5], BACF [56], SiamCAR [3], SiamRPN [2], MCCT [57], SRDCF [58], SwinTrack [50], and fDSST [59]. The Trans-HST is tested on HSV. The other trackers are tested on false-color videos and RGB videos because the inputs of the trackers include three channels. The false-color images are obtained by converting the HSV into three-channel videos [8]. The RGB videos are shot in the same scenes as the HSV. The success plots and the precision plots of the Trans-HST on HSV and the other trackers on false-color videos are shown in Fig. 6 and that of the Trans-HST on HSV and the other trackers on RGB videos are shown in Fig. 7.

As shown in Figs. 6 and 7, TransT achieves the best performance of RGB trackers on false-color videos and RGB videos and surpasses the second best trackers significantly. The performances of the RGB trackers on RGB videos are better than that on false-color images, which indicates that processing the false-color videos directly with the model trained with RGB data is irrational. The RGB trackers need to be transferred to the HS object tracking task for better performance. Accordingly, Trans-HST is built with the components

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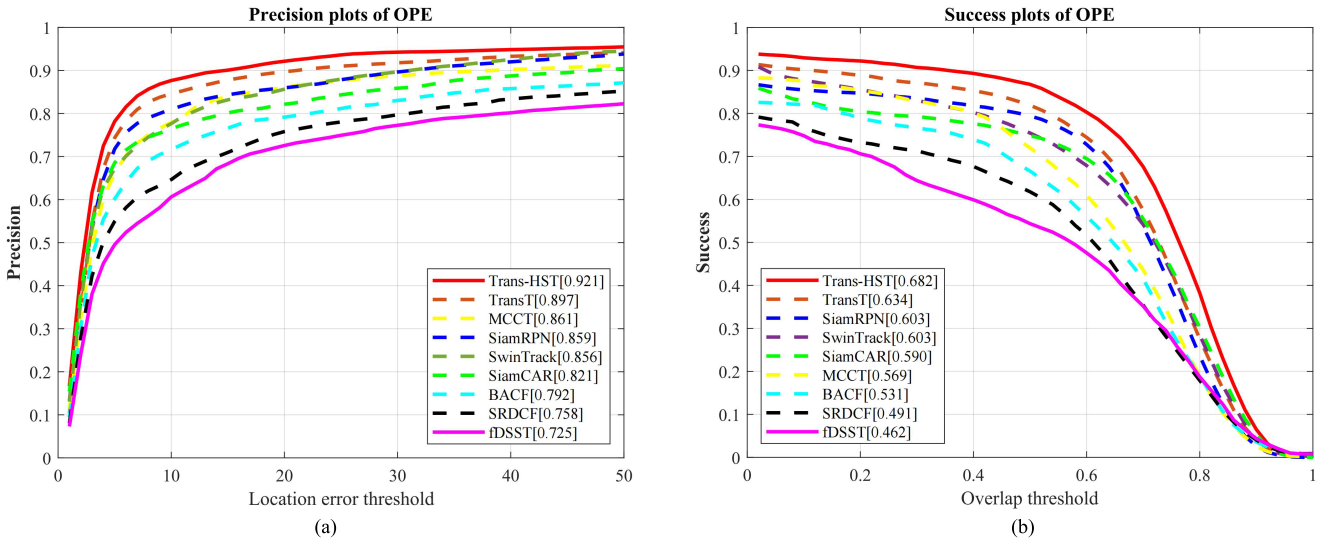


Fig. 7. Comparisons of Trans-HST and trackers on the corresponding RGB videos. (a) Precision plots. (b) Success plots.

TABLE III
PERFORMANCE COMPARISON WITH HYPERSPECTRAL TRACKERS OF AUC

| Trackers | Trans-HST | MHT | BAE-Net | SST-Net | DeepHKCF | MFI-HVT | SiamHT | SiamHYPER | SSDT-Net |
|---|---|---|---|---|---|---|---|---|---|
| Hyperspectral | 0.682 | 0.588 | 0.606 | 0.623 | 0.328 | 0.604 | 0.621 | 0.678 | 0.639 |

The top two values are marked by red and blue.

TABLE IV
ATTRIBUTE-BASED COMPARISON ON HSVS OR THE CORRESPONDING FALSE-COLOR VIDEOS

| Attributes | Trans-HST | SiamCAR | SiamRPN | SSDT_Net | SiamHYPER_BAN | BAE-Net | MHT |
|---|---|---|---|---|---|---|---|
| Background Clutter (BC) | 0.697 | 0.530 | 0.548 | 0.663 | 0.702 | 0.631 | 0.594 |
| Deformation (DEF) | 0.759 | 0.719 | 0.710 | 0.685 | 0.720 | 0.679 | 0.664 |
| Fast Motion (FM) | 0.673 | 0.665 | 0.612 | 0.603 | 0.710 | 0.607 | 0.541 |
| In-Plane Rotation (IPR) | 0.718 | 0.631 | 0.603 | 0.666 | 0.720 | 0.699 | 0.670 |
| Illumination Variation (IV) | 0.610 | 0.417 | 0.398 | 0.543 | 0.592 | 0.440 | 0.474 |
| Low Resolution (LR) | 0.691 | 0.435 | 0.429 | 0.521 | 0.664 | 0.491 | 0.478 |
| Motion Blur (MB) | 0.754 | 0.627 | 0.605 | 0.579 | 0.750 | 0.594 | 0.560 |
| Occlusion (OCC) | 0.630 | 0.527 | 0.499 | 0.607 | 0.635 | 0.555 | 0.565 |
| Out-Plane Rotation (OPR) | 0.724 | 0.654 | 0.635 | 0.695 | 0.706 | 0.693 | 0.631 |
| Out-of-View (OV) | 0.679 | 0.602 | 0.690 | 0.732 | 0.596 | 0.516 | 0.620 |
| Scale Variation (SV) | 0.672 | 0.548 | 0.543 | 0.639 | 0.657 | 0.608 | 0.564 |

The top two values are marked by red and blue.

of the transferred TransT, i.e., the feature extraction network, template-search features' fusion, and head network. With the improvement gained by the CBFE and TDFF, Trans-HST achieves the best performance in the two comparisons. Moreover, Trans-HST improves the performance of TransT by 7.3% and 4.8% for the AUC scores on false-color videos and RGB videos, respectively.

### *D. Quantitative Comparison With Hyperspectral Trackers*

The comparison between Trans-HST and recently released hyperspectral trackers is conducted to reveal the advancement of the proposed algorithm. The hyperspectral trackers include: MHT [7], BAE-Net [21], SST-Net [22], DeepHKCF [13], MFI-HVT [14], SiamHT [15], SiamHYPER [8], and SSDT-Net [60], and the results are reported in Table III. The trackers which convert the HSV into three-channel videos, i.e., MHT, DeepHKCF, MFI-HVT, SiamHT, and SSDT-Net, show weak performances. These trackers exploit different band selection or band fusion strategies to reserve the spectral information that have the strongest representational abilities and discard the other information. Nevertheless, the three channels of the converted videos contain insufficient spectral information for tracking the target in complex scenarios. The trackers, i.e., BAE-Net and SST-Net, divide the HSI into five false-color images and estimate the state of the targets by fusing the tracking results of the weak trackers which set the false-color images as inputs. Compared with the trackers which convert the HSI into three-channel false-color images, BAE-Net and SST-Net reserve most spectral information in the HSI, and only one band discarded for the HSI can be divided into five false-color images. However, the performances of

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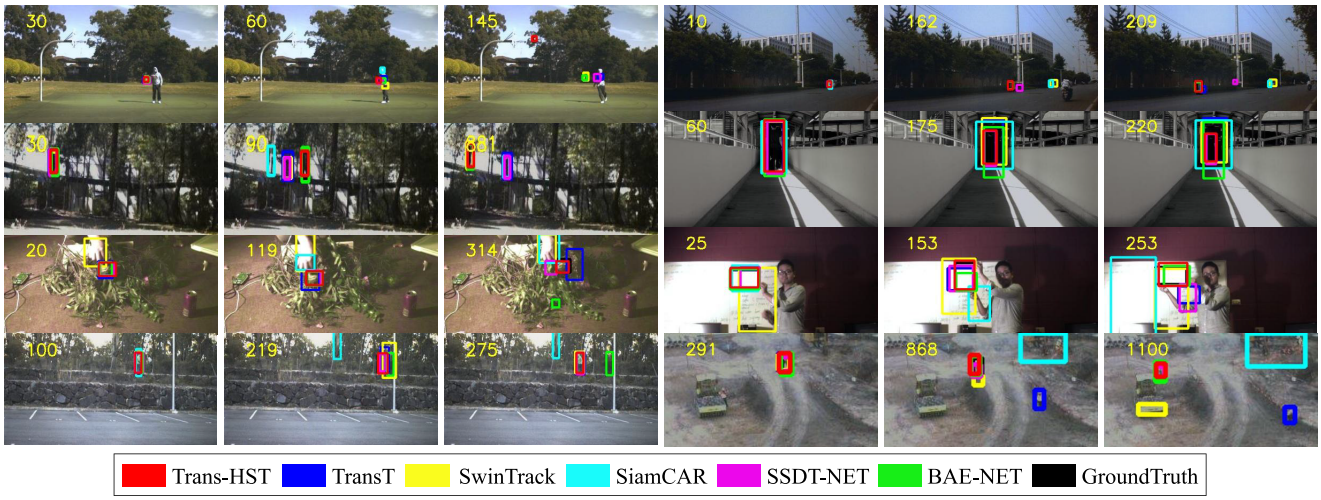


Fig. 8. Visualization results of our proposed hyperspectral tracker and several color trackers.

the two types of trackers are similar. Trans-HST also divides the HSI into five false-color images, but achieves significant better performance than BAE-Net and SST-Net. Trans-HST surpasses SST-Net by 5.9%. The CBFE and TDFF capture the long-range interaction information across the deep features corresponding to the false-color images and reduce the negative effects caused by the dividing process. SiamHYPER achieves the second AUC scores in this experiment because it applies HSV and RGB videos together to track the targets. Still, Trans-HST outperforms SiamHYPER and achieves the best AUC score, 0.682, in this comparison.

### *E. Attribute-Based Evaluation*

In this part, the AUC scores of the Trans-HST, two RGB trackers (SiamCAR and SiamRPN), and four hypersepectral trackers (SSDT_Net, SiamHYPER_BAN, BAE-Net and MHT) in 11 attributes are reported in Table IV. The RGB trackers are evaluated on the corresponding false-color videos. The hyperspectral trackers achieve better performances than that of the RGB trackers in BC and OCC, which indicates the advantages of HSV in the scenarios. The Trans-HST achieves the second performance in the two attributes, and the scores are close to the first. In the rest attributes, the Trans-HST perform the best for six out of nine attributes. Compared with SiamHYPER_BAN which tracks the target with RGB videos and HSV, the Trans-HST has a better performance for seven out of 11 attributes.

### *F. Visualizations of Tracking Results*

The tracking results of Trans-HST and five other tackers are visualized in Fig. 8. The trackers applied for comparison include three RGB trackers and two hyperspectral trackers. The sequences in Fig. 8, from top to bottom and left to right, are Basketball, Campus, Fruit, Forest2, Rider2, Student, Paper, and Worker. The eight sequences are hard for the three RGB trackers, i.e., TransT, SwinTrack, and SiamCAR, to track the target accurately, due to the background clutter, occlusion, and illumination variation due to the inherent limitation of the RGB videos. SSDT-NET is interrupted by an object which blocks the target and fails to track the target in Campus and Rider2. The bands' fusion strategy in SSDT-NET causes loss of the spectral information and reduces the discriminative ability of the tracker. Trans-HST can track the target accurately in all the nine sequences, which verifies that Trans-HST effectively exploits the spectral information against occlusion, background clutter, and illumination variation.

### *G. Limitations of Trans-HST*

The limitations of Trans-HST are summarized as follows, which are also considered as our future research orientation. First, the adaptively band regrouping method is not implemented in the Trans-HST. Bands in the HSI contribute to discriminating the target from the background differently in different scenarios. Therefore, some works have explored adaptively band regrouping method and achieved performance improvement [23], [61]. The Trans-HST can be further improved by adopting adaptively band regrouping method. Second, the speed of the Trans-HST does not reach real-time, which limits its application scenario. The limitations of the efficiency improvement are the feature extractions of five false-color images and learning the self-attention on the features with large number of channels. Exploring band selection method to reduce the number of the false-color images and channel reduction methods to reduce computational cost of the self-attention learning can be beneficial for the efficiency of the tracker.

## V. Conclusion

In this article, a new hyperspectral object tracking algorithm, Trans-HST, has been proposed. Trans-HST divides the HSI into multiple false-color images, and tracking the hyperspectral objects with the fused features corresponding to the false-color images. To decrease the interaction information loss caused by the dividing process, a feature enhancement module based on transformer, CBFE, is introduced. In addition, a new multiple feature fusion network, TDFF, is designed to fuse the deep features extracted from the false-color images and to augment the deep sematic information corresponding the original HSI

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into the fused features. The ablation study verified that the two proposed modules bring significant performance improvement. The comparison between the Trans-HST and other state-of-art RGB trackers and hyperspectral trackers reveals the superiority of the Trans-HST. Specifically, the Trans-HST outperforms the TransT, which is used to build and initialize the Trans-HST, by 7.3% and 4.8% for the AUC scores on false-color videos and RGB videos, respectively.

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